Climbing and Sliding Numbers

Questions

1. Complete the number tracks and describe the pattern.

a) 6 → 16 → 36 → 46 → 66 → ____

b) 248 → 228 → 208 → 188 →____

c) 820 → 920 → 1,020 → ____

→ **What operation is happening each time?**

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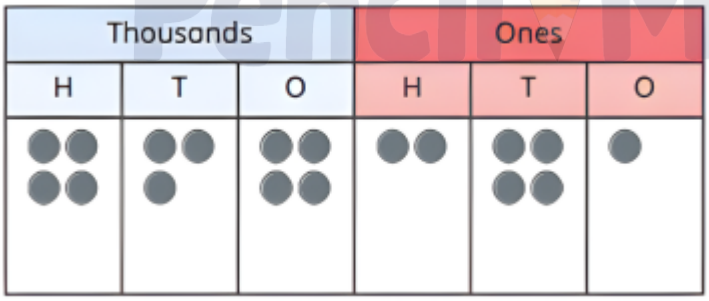
2. Aryan has completed this number track:

480 → 490 → 500 → 510 → 6,000 → 6,010

→ Spot and explain the mistake Aryan made.

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3. The number 434,241 is shown on a place value chart.



a) Arjun wants to find **1,000** more than **434,241.**

Explain why Arjun's method would work.

What is **1,000** more than **434,241?**

b) Find **100** less than **434,241**.

c) Find **10** more than **434,241.**

d) Find **1** less than **434,241.**

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4. A number is shown on a Gattegno chart.

100,000	200,000	300,000	400,000	500,000	600,000	700,000	800,000	900,000
10,000	20,000	30,000	40,000	50,000	60,000	70,000	80,000	90,000
1,000	2,000	3,000	4,000	5,000	6,000	7,000	8,000	9,000
100	200	300	400	500	600	700	800	900
10	20	30	40	50	60	70	80	90
1	2	3	4	5	6	7	8	9

a) What number is represented?
b) What is **1,000** more than this number? How do you know?
c) What is **10,000** less than this number? How do you know?
d) What happens when a counter moves beyond the highest number its row?

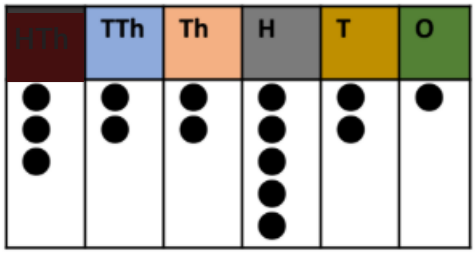
5. Complete the table.

Number	10 more	100 more	1,000 more	10,000 more	100,000 more
32					
370					
3,700					
37,000					

What pattern do you notice across the rows?

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6. Liam uses a place value chart.



He adds two counters to the tens column.

a) What new number could Liam create?

b) What happens if he adds two counters to the thousands column instead?

7. Amira says, "When you subtract 10,000 from 250,000, only the ten-thousands place changes."

a) Do you agree with Amira?

b) Explain your reasoning with an example.

8. Lia says, "If I add 1,000 to 56,789, only the thousands digit will change."

a) Is Lia correct?

b) Explain why or why not→ Write it in words: _____

9. Charlie says, "If you add 100,000 to any 5-digit number, the tens digit always changes."

a) Is Charlie's statement always true, sometimes true, or never true?

b) Give an example to prove your answer.

10. A number increases by 10,000 and becomes 483,214.

a) What could the original number have been?

b) Explain how you worked it out.